

Lantern Pharma, Inc.

(LTRN – \$1.24)

Buy

Price Target: \$25

Chad Messer, Ph.D.

Senior Research Analyst

612-294-6913

chad.messer@lakestreetcm.com

Reports 4Q:2025, Seeking Changes To HARMONIC Protocol To Speed Up Enrollment

Type C Meeting With FDA Scheduled For Mid-May To Address Enrollment Issues With HARMONIC – Lantern recently submitted a Type C meeting package to the FDA regarding the Phase 2 HARMONIC trial for LP-300 in never-smokers with NSCLC. The meeting, scheduled for mid-May 2026, seeks concurrence on three major protocol amendments: (i) focusing future enrollment on patients with the EGFR exon 21 L858R mutation; (ii) increasing the maximum treatment cycles from six to eight; and (iii) converting the study from a randomized study to a single-arm Simon two-stage design. Focusing enrollment on L858R mutants is supported by preliminary data suggesting higher clinical benefit in this specific mutation subtype. The switch to a single arm Simon 2 stage design was meant to mitigate challenges by enrolling patients into an active control arm using only chemotherapy given the emergence of additional treatment options. As a reminder, enrollment in HARMONIC was expanded to Asia last year, and enrollment in the Japanese cohort was completed ahead of schedule in July.

withZeta.ai Represents Near-Term AI Revenue Opportunity – In late 2025, Lantern introduced withZeta.ai, a multi-agentic AI co-scientist platform focused on rare cancers. Since December 2025, the platform has been in beta testing with over 25 biotech companies and research centers. Management expects to provide updates on commercial traction and subscription-based partnerships throughout 2026. To support this and the broader RADR platform, Lantern initiated an AI Center of Excellence in India in early 2026 to industrialize these capabilities.

Clinical Pipeline Advancing Toward Phase 1b/2 Milestones –

- **LP-184:** Phase 1a completed with all primary endpoints met and a 48% clinical benefit rate at or above therapeutic dose thresholds. Lantern plans to initiate biomarker-guided Phase 1b/2 trials in TNBC and NSCLC in H1 2026, subject to funding.
- **Starlight Therapeutics:** The FDA cleared the IND for a Phase 1 pediatric CNS cancer trial of STAR-001 (LP-184) in early 2026. The program has received Rare Pediatric Disease and Orphan Drug Designations, which could qualify Lantern for a Priority Review Voucher (PRV) upon approval.
- **LP-284:** Received FDA Orphan Drug Designation for soft tissue sarcomas in Q1 2026.

Financial Position – Lantern ended 2025 with ~\$10.1 million in cash. Total operating expenses for FY 2025 were \$18.0 million, a 19% reduction year-over-year from \$22.2 million in FY 2024. This decrease was largely driven by a 29% reduction in R&D expenses due to efficiencies in research studies and materials. Management projects current cash will fund operations until approximately late July to mid-September 2026 and is actively evaluating options for continued funding of the company. During 2025, the company sold 356,922 shares under its ATM agreement for gross proceeds of ~\$1.6 million.

Changes	Previous	Current
Rating:	--	Buy
Price Target:	--	\$25
FY25A Rev (M):	--	\$0.0
FY26E Rev (M):	--	\$0.0
FY25A EPS:	(\$1.67)	(\$1.57)
FY26E EPS:	(\$1.91)	(\$1.57)

Profile

Price:	\$1.24
Shares Out (M):	11.2
Market Cap (M):	\$14
Enterprise Value (M):	(\$2)
Avg Daily Vol (K):	3,148
Insiders Own:	24.1%
Book Val / Sh:	\$5.70
Net Cash / Sh:	\$1.42
Div / Yield:	NA

Rev (M)	FY24A	FY25A	FY26E
Mar	\$0.0	\$0.0	--
Jun	\$0.0	\$0.0	--
Sep	\$0.0	\$0.0	--
Dec	\$0.0	\$0.0	--
FY	\$0.0	\$0.0	\$0.0

EPS	FY24A	FY25A	FY26E
Mar	(\$0.51)	(\$0.42)	--
Jun	(\$0.46)	(\$0.40)	--
Sep	(\$0.42)	(\$0.39)	--
Dec	(\$0.54)	(\$0.36)	--
FY	(\$1.93)	(\$1.57)	(\$1.57)

Valuation	FY24A	FY25E	FY26E
EV/Sales	NM	NM	NM
P/E	NM	NM	NM

Management

CEO	Panna Sharma
CFO	David Margrave

Company Description

Lantern Pharma is a clinical-stage biopharma company that leverages artificial intelligence (AI), machine learning, and genomic data to discover, develop, and personalize precision oncology therapies, with a focus on improving the success rates and efficiency of cancer drug development.

VALUATION

Price Target – Our \$25 target is based exclusively on LP-300, leaving LP-184, LP-284, the ADC programs, and partnered programs as upside. For the US market, we apply a standard 5x multiple to our 2032 sales forecast of \$706M and discount back 7 years to YE:25 by 50% annually. We believe this high discount rate reflects the risks associated with the Phase 2 development status of the drug and the fact that current development is attempting to prove *a post hoc* hypothesis from prior failed trials. For Europe, we apply the same 5x multiple to our 2032 sales forecast of \$403M and use the same 50% discount rate. As a proxy for the Asian market, we forecast sales of \$149M in 2032 in Japan and South Korea, again using a 5x multiple and 50% discount rate. See below for further details on our sales forecasts for LP-300 by region. We further assume Lantern will pay an aggregate 20% in royalties, yielding a \$294M enterprise value. We then add our estimate of YE:25 cash of \$7.0M and divide by an estimated 10.9M shares outstanding and account for the exercise of 1.2M shares in outstanding options to arrive at our \$25 price target.

Upcoming Milestones

- **HARMONIC Type C Meeting with FDA (May 2026)**

RISKS

We believe an investment in Lantern involves certain general risks typical of investing in development-stage biotechnology companies, as well as company-specific risks. These include:

- **Clinical** – Lantern’s clinical programs, including the ongoing Phase 2 study of LP-300 and the Phase 1 studies of LP-184 and LP-284, may fail. LP-300 failed in 5 prior Phase 3 studies, and although 2 of those studies yielded encouraging evidence of efficacy in the non-smoking population of lung cancer patients, that signal may not repeat in a prospective study.
- **Regulatory** – Lantern has not received regulatory approval from any agency for any of its products. Lantern’s products, including LP-184, LP-284, and LP-300, will all require approval from regulatory agencies, including the U.S. FDA, in order to reach the market. Even if the clinical programs for one or more of these programs meet their objectives, there is no guarantee that the drugs will get approved as regulatory review considers multiple factors in addition to clinical trial success.
- **Commercial** – All of Lantern’s products, if approved, will compete with numerous existing and potential future existing products. While no product is specifically approved for non-smokers in lung cancer, there are dozens of approved drugs for lung cancer, including TKIs, immune checkpoint inhibitors, and chemotherapy. Even if LP-300 receives approval in never smoker lung cancer patients, there is no guarantee that it can win market shares and achieve commercial success.
- **Financial** – Lantern ended 2025 with \$10.1M in cash, which we forecast will run out well before any potential revenues from drug sales. Lantern will likely have to raise additional capital in order to finish clinical development of LP-300 or any of its other programs. There is no guarantee that it will be able to obtain that capital on reasonable terms or at all.

FINANCIAL MODEL

Lantern Pharma

Fiscal Year End Dec 30	2023	2024					2025					2026					2027	2028	2029	2030
Income Statement	2023A	Mar-24A	Jun-24A	Sep-24A	Dec-24A	2024A	Mar-25A	Jun-25A	Sep-25A	Dec-25A	2025A	Mar-26E	Jun-26E	Sep-26E	Dec-26E	2026E	2027E	2028E	2029E	2030E
<i>(\$ in millions, except per unit amounts)</i>		Q1	Q2	Q3	Q4		Q1	Q2	Q3	Q4		Q1	Q2	Q3	Q4					
Operating expenses:																				
G&A	5,983,255	1,481,215	1,519,724	1,462,930	1,626,878	6,090,747	1,510,077	1,583,521	1,912,829	1,457,944	6,464,371	1,500,000	1,500,000	1,500,000	2,000,000	6,500,000	7,250,000	7,500,000	7,750,000	8,000,000
R&D	11,894,315	4,250,786	3,888,737	3,716,646	4,269,521	16,125,690	3,263,955	3,068,379	2,436,971	2,744,818	11,514,123	2,750,000	3,000,000	3,000,000	3,250,000	12,000,000	16,000,000	17,000,000	18,000,000	20,000,000
Total operating expenses	17,877,570	5,732,001	5,408,461	5,179,576	5,896,399	22,216,437	4,774,032	4,651,900	4,349,800	4,202,762	17,978,494	4,250,000	4,500,000	4,500,000	5,250,000	18,500,000	23,250,000	24,500,000	25,750,000	28,000,000
Loss from Operations	(17,877,570)	(5,732,001)	(5,408,461)	(5,179,576)	(5,896,399)	(22,216,437)	(4,774,032)	(4,651,900)	(4,349,800)	(4,202,762)	(17,978,494)	(4,250,000)	(4,500,000)	(4,500,000)	(5,250,000)	(18,500,000)	(23,250,000)	(24,500,000)	(25,750,000)	(28,000,000)
Interest income	765,388	200,950	188,660	191,352	161,393	742,355	149,790	114,745	100,921	72,475	437,931	60,000	45,000	30,000	15,000	150,000	100,000	100,000	100,000	100,000
Other Income	1,150,648	90,241	260,295	482,527	(140,194)	692,869	87,459	206,140	71,456	56,070	421,125	100,000	100,000	100,000	100,000	400,000	400,000	400,000	400,000	400,000
Net income (loss)	(15,961,534)	(5,440,810)	(4,989,506)	(4,505,697)	(5,875,200)	(20,781,213)	(4,536,783)	(4,331,015)	(4,177,423)	(4,074,217)	(17,119,438)	(4,090,000)	(4,355,000)	(4,370,000)	(5,135,000)	(17,950,000)	(22,750,000)	(24,000,000)	(25,250,000)	(27,500,000)
GAAP Basic EPS	(1.47)	(0.51)	(0.46)	(0.42)	(0.54)	(1.93)	(0.42)	(0.40)	(0.39)	(0.36)	(1.57)	(0.36)	(0.38)	(0.38)	(0.44)	(1.57)	(2.02)	(2.11)	(2.20)	(2.38)
Weighted average shares outstanding	10,842,365	10,742,797	10,758,805	10,763,351	10,784,323	10,762,319	10,784,725	10,784,725	10,833,393	11,189,857	10,898,175	11,260,000	11,360,000	11,460,000	11,560,000	11,410,000	11,264,857	11,364,857	11,464,857	11,564,857
Lake Street Capital Markets, LLC Chad Messer, Ph.D. 3/31/2026																				

IMPORTANT DISCLOSURES



Initiate: April 2, 2025 – Rating: Buy - Price Target: \$25

RATINGS DEFINITION

Buy rated stocks are expected to generate greater than 10% returns during the next 12 months. **Hold** rated stocks are expected to generate returns of 0% to 10% during the next 12 months. **Sell** rated stocks are expected to generate negative returns over the next 12 months and generally do not have a price target.

Information on our valuation methodology and risks can be found in the “Investment Thesis & Valuation” and “Risks” sections above.

RATINGS DISTRIBUTION

(as of March 13, 2026)

Rating	All Covered Companies (%)	Investment Banking Clients (%)
Buy	94.8%	23.8%
Hold	5.2%	10.0%
Sell	0.0%	0.0%

RESEARCH DISCLOSURES

Lake Street Capital Markets, or its affiliates, intends to seek or expects to receive compensation for investment banking services from the subject issuer in the next three months. The authoring analysts who are responsible for the preparation of this investment research are eligible for compensation based on the total revenue and general profitability of Lake Street Capital Markets, which includes investment banking revenue. However, such authoring analyst will not receive compensation that is directly based on or linked to specific investment banking transactions.

ANALYST CERTIFICATION – REGULATION AC

Each authoring analyst of Lake Street Capital Markets whose name appears on the front page of this investment research hereby certifies that (1) the recommendations and opinions expressed in this investment research accurately reflect the authoring analyst’s personal, independent, and objective views about any and all of the subject investments or issuers discussed herein; and (2) no part of the authoring analyst’s compensation was, is, or will be, directly or indirectly, related to the specific recommendations or views expressed by the authoring analysts.

OTHER DISCLOSURES

This investment research is intended for informational purposes only and does not constitute an offer or solicitation to buy or sell securities referenced herein in any jurisdiction where such would be prohibited. Although the information and statements of fact contained in this investment research have been obtained from sources we believe to be reliable, including statistical services and issuer statements & reports, its accuracy and completeness cannot be guaranteed. The securities discussed and any recommendations made herein may not be suitable for all investors depending on their specific financial position or investment objectives. This investment research and any opinions, recommendations, and estimates contained here represent Lake Street Capital Market’s judgment as of the date of this report and are subject to change without notice. Lake Street Capital Markets may effect transactions as a principal or agent in the securities mentioned herein. Additional information available on request. Member FINRA / SIPC. This report may not be reproduced, distributed, or published without the prior consent of Lake Street Capital Markets, LLC. All rights reserved.